

Job No. exp no 08	Sheet No. 1	Rev
Job Title ANALYSIS AND DESIGN ANY PORTAL FRAME WITH SWAY	Part	Ref
Client	By P.GOWTHAMI	Date 9/15/2026
File ANALYSIS AND DESIGN ANY PORTAL FRAME WITH SWAY.std	Date Time 15-Sep-2026 14:46	Chd

Job Information

	Engineer	Checked	Approved
Name:	P.GOWTHAMI		
Date:	9/15/2026		

Comments:	
Structure Type:	SPACE FRAME

Geometry

Entity Type	Count	Highest
Nodes	4	4
Analytical Members	3	3

Load Cases

Load Case Type	Count
Primary	2
Combination	1

Included in this printout are data for:

All	The Whole Structure
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Load Case Table

Included in this printout are results for load cases:

L/C	Type	Name
1	Primary	DEAD LOAD
2	Primary	LIVE LOAD
3	Combination	COMBINATION LOAD CASE 3

Nodes

Node	X (m)	Y (m)	Z (m)
1	0.000	0.000	0.000
2	5.000	0.000	0.000
3	0.000	4.000	0.000
4	5.000	4.000	0.000

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Beams

Beam	Node A	Node B	Length (m)	Property	β (degrees)
1	1	3	4.000	1	0.000
2	3	4	5.000	1	0.000
3	4	2	4.000	1	0.000

Supports

Node	X (kip/in)	Y (kip/in)	Z (kip/in)	rX (kip-ft/deg)	rY (kip-ft/deg)	rZ (kip-ft/deg)
1	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
2	Fixed	Fixed	Fixed	-	-	-

Combination Load Cases

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
3	COMBINATION LOAD CASE 3	1	DEAD LOAD	1.500
		2	LIVE LOAD	1.500

Beam Displacement Summary

Type	Beam	L/C	d (m)	X (in)	Y (in)	Z (in)	Resultant (in)
Max X	3	3	1.000	0.129	-0.003	-1.830	1.835
Min X	2	1	2.500	0.000	0.000	0.000	0.000
Max Y	3	3	4.000	0.000	0.000	0.000	0.000
Min Y	3	3	0.000	0.127	-0.004	-2.350	2.353
Max Z	3	3	4.000	0.000	0.000	0.000	0.000
Min Z	3	3	0.000	0.127	-0.004	-2.350	2.353
Max Rst	2	3	5.000	0.127	-0.004	-2.350	2.353

Beam End Force Summary

	Beam	Node	L/C	Axial Fx (kN)	Shear Fy (kN)	Fz (kN)	Torsion Mx (kip-in)	Bending My (kip-in)	Mz (kip-in)
Max Fx	3	4	3	89.902	-5.255	-14.395	796.567	509.616	610.539
Min Fx	2	3	3	-5.255	60.098	-14.395	509.616	159.547	-48.927

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Beam End Force Summary Cont...

	Beam	Node	L/C	Axial	Shear		Torsion	Bending	
				Fx (kN)	Fy (kN)	Fz (kN)	Mx (kip-in)	My (kip-in)	Mz (kip-in)
Max Fy	2	3	3	-5.255	60.098	-14.395	509.616	159.547	-48.927
Min Fy	2	4	3	-5.255	-89.902	-14.395	509.616	796.567	610.539
Max Fz	1	1	1	0.000	0.000	0.000	0.000	0.000	0.000
Min Fz	1	1	3	60.098	50.255	-14.395	956.114	0.000	933.672
Max Mx	3	4	3	89.902	-5.255	-14.395	796.567	509.616	610.539
Min Mx	1	1	3	60.098	50.255	-14.395	956.114	0.000	933.672
Max My	3	4	3	89.902	-5.255	-14.395	796.567	509.616	610.539
Min My	2	4	3	-5.255	-89.902	-14.395	509.616	796.567	610.539
Max Mz	1	1	3	60.098	50.255	-14.395	956.114	0.000	933.672
Min Mz	1	3	3	60.098	5.255	-14.395	159.547	509.616	-48.927

Beam Force Detail Summary

	Beam	L/C	d (m)	Axial	Shear		Torsion	Bending	
				Fx (kN)	Fy (kN)	Fz (kN)	Mx (kip-in)	My (kip-in)	Mz (kip-in)
Max Fx	3	3	0.000	89.902	-5.255	-14.395	796.567	509.616	610.539
Min Fx	2	3	0.000	-5.255	60.098	-14.395	509.616	-159.547	-48.927
Max Fy	2	3	0.000	-5.255	60.098	-14.395	509.616	-159.547	-48.927
Min Fy	2	3	5.000	-5.255	-89.902	-14.395	509.616	-796.567	610.539
Max Fz	1	1	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Min Fz	1	3	0.000	60.098	50.255	-14.395	-956.114	0.000	933.672
Max Mx	3	3	0.000	89.902	-5.255	-14.395	796.567	509.616	610.539
Min Mx	1	3	0.000	60.098	50.255	-14.395	-956.114	0.000	933.672
Max My	3	3	0.000	89.902	-5.255	-14.395	796.567	509.616	610.539
Min My	2	3	5.000	-5.255	-89.902	-14.395	509.616	-796.567	610.539
Max Mz	1	3	0.000	60.098	50.255	-14.395	-956.114	0.000	933.672
Min Mz	2	3	2.500	-5.255	-14.902	-14.395	509.616	-478.057	-548.955

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Beam Max Moments

Beam	Node A	Length (m)	L/C		d (m)	Max My (kip-in)	d (m)	Max Mz (kip-in)
1	1	4.000	1	Max +ve	0.000	0.000	0.000	0.000
				Max -ve	0.000	0.000	0.000	0.000
			2	Max +ve	0.000	0.000	0.000	622.448
				Max -ve	4.000	-339.744	4.000	-32.618
			3	Max +ve	0.000	0.000	0.000	933.672
				Max -ve	4.000	-509.616	4.000	-48.927
2	3	5.000	1	Max +ve	0.000	0.000	0.000	0.000
				Max -ve	0.000	0.000	0.000	0.000
			2	Max +ve	0.000	0.000	5.000	407.026
				Max -ve	5.000	-531.045	2.083	-387.241
			3	Max +ve	0.000	0.000	5.000	610.539
				Max -ve	5.000	-796.567	2.083	-580.861
3	4	4.000	1	Max +ve	0.000	0.000	0.000	0.000
				Max -ve	0.000	0.000	0.000	0.000
			2	Max +ve	0.000	339.744	2.000	469.037
				Max -ve	4.000	0.000	0.000	0.000
			3	Max +ve	0.000	509.616	2.000	703.555
				Max -ve	4.000	0.000	0.000	0.000

Beam Combined Stresses Summary

Beam	L/C	Length (m)	Max Comp			Max Tens		
			Stress (psi)	d (m)	Corner	Stress (psi)	d (m)	Corner
1	1	4.000	0.000	0.000	0	0.000	0.000	0
	2	4.000	680.584	4.000	1	616.017	4.000	2
	3	4.000	1,020.876	4.000	1	924.025	4.000	2
2	1	5.000	0.000	0.000	0	0.000	0.000	0
	2	5.000	1,334.657	5.000	4	1,340.302	5.000	2

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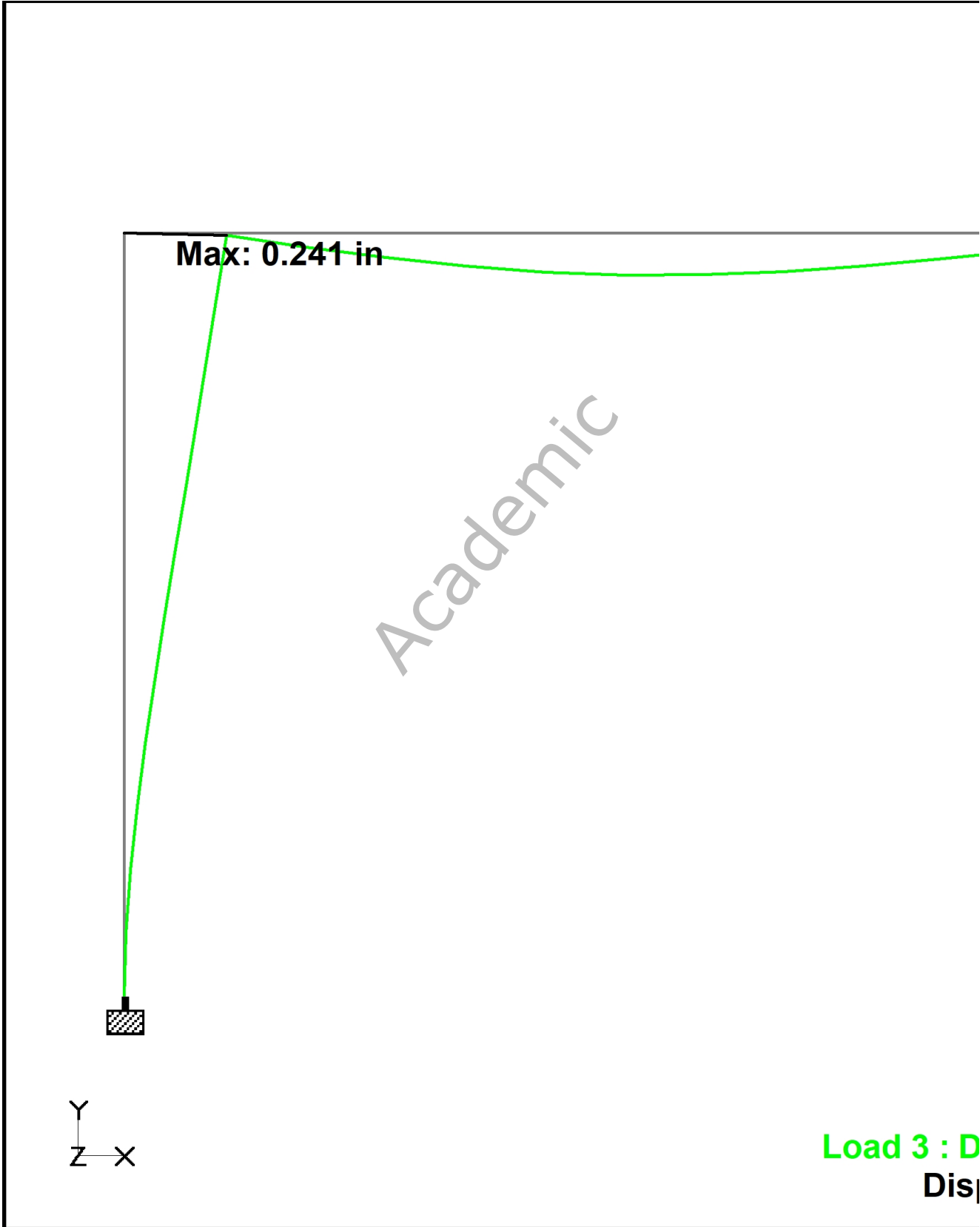
Beam Combined Stresses Summary Cont...

Beam	L/C	Length (m)	Max Comp			Max Tens		
			Stress (psi)	d (m)	Corner	Stress (psi)	d (m)	Corner
2	3	5.000	2,001.9 85	5.000	4	- 2,010.4 53	5.000	2
3	1	4.000	0.000	0.000	0	0.000	0.000	0
	2	4.000	1,037.4 54	0.000	2	- 940.867	0.000	1
	3	4.000	1,556.1 81	0.000	2	- 1,411.3 00	0.000	1

Node Displacement Summary

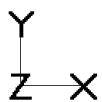
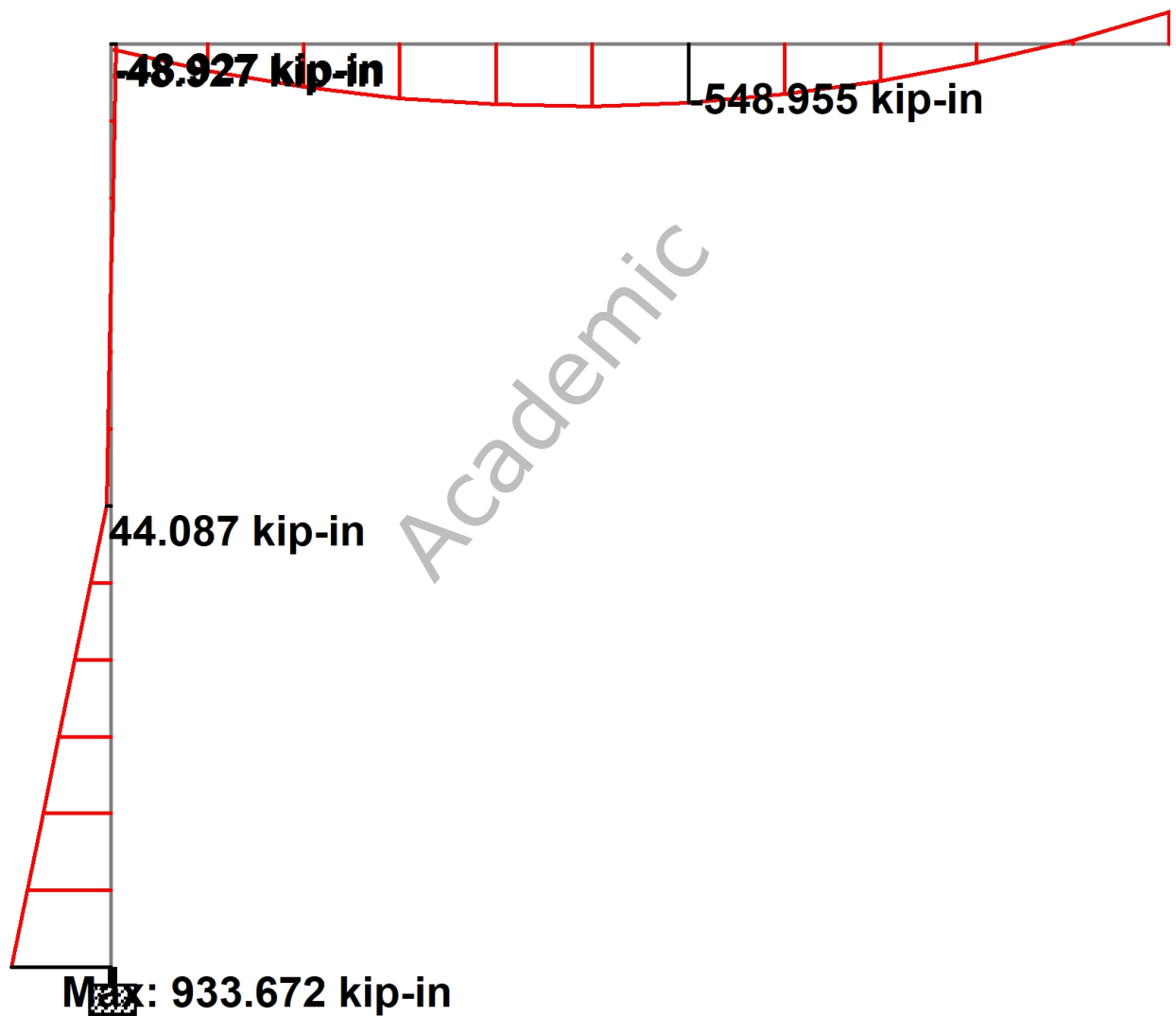
Type	Node	L/C	X (in)	Y (in)	Z (in)	Result ant (in)	rX (rad)	rY (rad)	rZ (rad)
Max X	4	3	0.127	-0.004	-2.350	2.353	-0.012	0.017	0.000
Min X	1	1	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Max Y	1	1	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Min Y	4	3	0.127	-0.004	-2.350	2.353	-0.012	0.017	0.000
Max Z	1	1	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Min Z	4	3	0.127	-0.004	-2.350	2.353	-0.012	0.017	0.000
Max rX	1	1	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Min rX	2	3	0.000	0.000	0.000	0.000	-0.016	0.022	-0.002
Max rY	2	3	0.000	0.000	0.000	0.000	-0.016	0.022	-0.002
Min rY	1	1	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Max rZ	4	3	0.127	-0.004	-2.350	2.353	-0.012	0.017	0.000
Min rZ	2	3	0.000	0.000	0.000	0.000	-0.016	0.022	-0.002
Max Rst	4	3	0.127	-0.004	-2.350	2.353	-0.012	0.017	0.000

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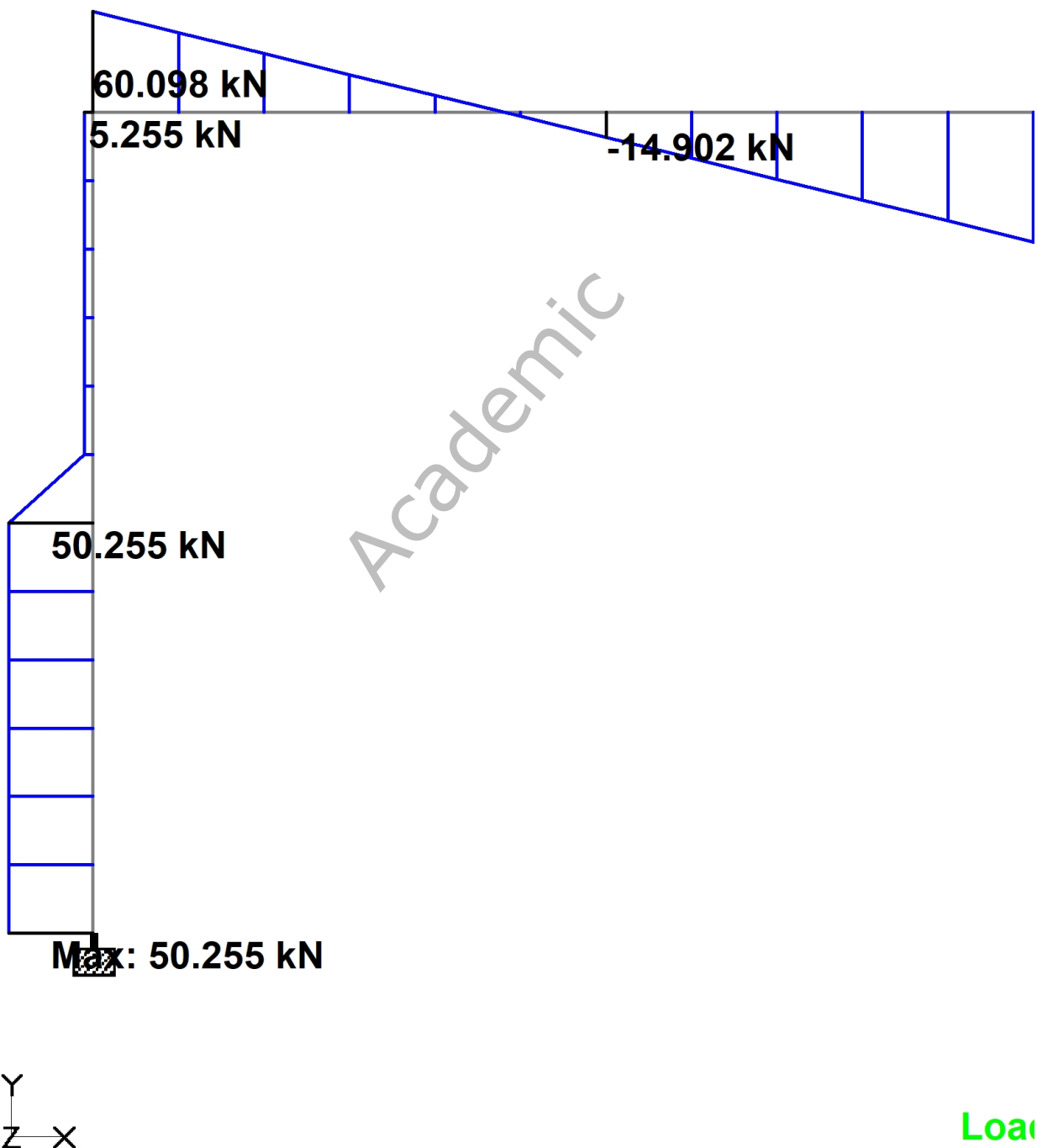
Whole Structure Displacements 0.23622in:1m 3 COMBINATION LOAD CASE 3

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Load 3
M

Whole Structure Mz 2165.35kip-in:1m 3 COMBINATION LOAD CASE 3



Whole Structure Fy 122.589kN:1m 3 COMBINATION LOAD CASE 3